Unified Second-Order Consolidation for Model Merging

From CRL to HTCL to RegMean, and Back

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Abstract

This note develops a single quadratic-surrogate framework that recovers three superficially distinct consolidation schemes as special cases: the Hessian-weighted averaging used in our Consolidated Reinforcement Learner (CRL), the sequential second-order hierarchical update in HTCL, and the Regression-Mean (RegMean) objective for dataless model merging. The derivation starts from a functional matching objective and linearizes either the loss or the model output around each expert's weights. The unifying object is a per-expert positive semidefinite metric whose particular choice recovers the loss Hessian H_i, the empirical Fisher F_i, the input Gram matrix $X_i^\top X_i$, or the Gauss-Newton kernel $J_i^\top J_i$ depending on modeling assumptions. We prove a Gauss-Newton identity that expresses the loss Hessian H_i as the sum of the Gauss-Newton kernel G_i and a residual term proportional to the output-loss gradient, and give a mixed closed-form solution that uses the full Hessian whenever it is available. We address two failure modes of the CRL formulation for reinforcement learning, namely (i) experts that are not trained to convergence, so $\nabla L_i(w_i^\star) \neq 0$, and (ii) expert optima that are far apart in parameter space, invalidating a single Taylor expansion. The functional-matching formulation removes failure mode (i) entirely and the mixed Hessian formulation quantifies when and why the loss Hessian should be used in preference to the Jacobian Gram. A regularized closed-form solution is given and the algorithmic implications for merging RL policies are discussed.

1 Setup

Let $\mathcal{W} = \mathbb{R}^d$ be a shared parameter space and let $\{(\mathcal{D}_i, w_i^\star)\}_{i=1}^N$ be a collection of expert models, where expert i has been trained on dataset $\mathcal{D}_i$ and has weights $w_i^\star$. Each expert exposes a predictor $f_{w_i^\star} : \mathcal{X} \to \mathbb{R}^o$ and a per-task loss $L_i : \mathcal{W} \to \mathbb{R}$. Nonnegative mixing weights $\alpha_i > 0$ with $\sum_i \alpha_i = 1$ reflect the relative importance of expert i.

Our goal is to produce a single merged weight $w^\star \in \mathcal{W}$ whose predictor $f_{w^\star}$ inherits the behavior of every expert as closely as possible on its own data distribution, without access to the raw data.

2 Two Families of Objective

There are two canonical ways to define what "close to every expert" means.

Loss-based objective. Minimize the α-weighted sum of the per-task losses,

$$\mathcal{L}_{\text{loss}}(w) \;=\; \sum_{i=1}^{N} \alpha_i \, L_i(w). \tag{1}$$

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This is the objective underlying the CRL derivation. Its natural quadratic approximation is the second-order Taylor expansion of each L_i around $w_i^\star$.

Function-matching objective. Minimize the α-weighted functional distance from each expert on its own dataset,

$$\mathcal{L}_{\text{fn}}(w) \;=\; \sum_{i=1}^{N} \alpha_i \, \mathbb{E}_{x\sim\mathcal{D}_i}\Big[\, \big\| f_w(x) - f_{w_i^\star}(x) \big\|^2 \Big]. \tag{2}$$

This is the objective underlying RegMean and, implicitly, policy distillation in reinforcement learning. Its natural quadratic approximation is the first-order Taylor expansion of $f_w(x)$ around $w_i^\star$.

The two objectives agree only for a restricted class of models and losses. In general they produce different curvature metrics and different handling of the non-convergence of experts. We treat them separately and then unify.

3 Derivation A: Loss-Based Consolidation

3.1 Ideal case (CRL)

Assume each expert reaches a stationary point of its task loss.
Assumption 1 (Expert optimality)**.** $\nabla L_i(w_i^\star) = 0$ *for every* i.

Taylor-expanding L_i around $w_i^\star$ to second order and substituting Assumption 1 gives

$$L_i(w) \;=\; L_i(w_i^\star) \;+\; \tfrac{1}{2}(w - w_i^\star)^\top H_i \,(w - w_i^\star) \;+\; R_i^{(3)}(w), \tag{3}$$

where $H_i = \nabla^2 L_i(w_i^\star) \succeq 0$ is the Hessian at the optimum and $R_i^{(3)}$ collects cubic and higher remainder terms. Substituting (3) into (1) and discarding constants in w as well as the remainder gives the quadratic surrogate

$$\widetilde{\mathcal{L}}_{\text{loss}}(w) \;=\; \tfrac{1}{2}\sum_{i=1}^{N} \alpha_i (w - w_i^\star)^\top H_i (w - w_i^\star). \tag{4}$$

Expanding and collecting terms of equal order in w yields

$$\widetilde{\mathcal{L}}_{\text{loss}}(w) \;=\; C \;+\; \tfrac{1}{2} w^\top H_{\text{agg}} w \;-\; w^\top b_{\text{agg}}, \tag{5}$$

with

$$H_{\text{agg}} \;=\; \sum_{i=1}^{N} \alpha_i H_i, \qquad b_{\text{agg}} \;=\; \sum_{i=1}^{N} \alpha_i H_i w_i^\star. \tag{6}$$

Stationarity $\nabla_w \widetilde{\mathcal{L}}_{\text{loss}} = 0$ gives the linear system $H_{\text{agg}} w = b_{\text{agg}}$ whose solution, when $H_{\text{agg}} \succ 0$, is the CRL formula

$$\boxed{w_{\text{CRL}}^\star \;=\; \left(\sum_{i=1}^{N} \alpha_i H_i\right)^{-1} \left(\sum_{i=1}^{N} \alpha_i H_i w_i^\star\right).} \tag{7}$$

3.2 Realistic case (no optimality)

Assumption 1 is fragile. In reinforcement learning the per-task objective is nonstationary, reward signals are stochastic, and the policy gradient rarely drives ∇L_i exactly to zero. Dropping Assumption 1 and retaining the first-order term in the Taylor expansion gives

$$L_i(w) \;\approx\; L_i(w_i^\star) \;+\; g_i^\top (w - w_i^\star) \;+\; \tfrac{1}{2}(w - w_i^\star)^\top H_i (w - w_i^\star), \qquad g_i = \nabla L_i(w_i^\star). \tag{8}$$

The stationary condition for $\mathcal{L}_{\text{loss}}$ becomes

$$\sum_{i=1}^{N} \alpha_i \big[\, H_i(w - w_i^\star) + g_i \,\big] \;=\; 0, \tag{9}$$

which after rearrangement yields the gradient-corrected estimate

$$\boxed{w_{\text{loss}}^\star \;=\; \left(\sum_i \alpha_i H_i\right)^{-1} \left(\sum_i \alpha_i H_i w_i^\star \;-\; \sum_i \alpha_i g_i\right).} \tag{10}$$

Equation (10) reduces to (7) when every $g_i = 0$. It is the correct formula whenever expert training is incomplete.

Remark 1 (Connection to HTCL)**.** *Our HTCL update in the two-level case,*

$$w_1^{(t)} \;=\; w_1^{(t-1)} \;+\; \Big(H^{(t-1)} + \lambda I\Big)^{-1} \big(\lambda\, \Delta d^{(t)} \;-\; g^{(t-1)}\big), \tag{11}$$

is exactly the one-step, two-expert specialization of (10) *augmented with a proximal pull* $\frac{\lambda}{2}\|w - w_\ell^{(t)}\|^2$ *toward the current local model. To see this, take* $N = 1$ *in* (10)*, identify* $w_1^\star \leftarrow w_1^{(t-1)}$*,* $H_1 \leftarrow H^{(t-1)}$*,* $g_1 \leftarrow g^{(t-1)}$*, and add the proximal penalty; the stationary condition then produces the HTCL update directly. HTCL therefore already implements the realistic, gradient-corrected consolidation at every level of its hierarchy.*

4 Derivation B: Function-Matching Consolidation

The loss-based derivation inherits two weaknesses. It requires that H_i be computable at $w_i^\star$, and it rests on Assumption 1 to eliminate the gradient term. Both concerns disappear if the objective is (2) instead of (1).

4.1 Linearization of the predictor

Let $J_i(x) = \nabla_w f_w(x)\big|_{w=w_i^\star} \in \mathbb{R}^{o\times d}$ denote the Jacobian of the predictor at the expert's weights. A first-order Taylor expansion gives

$$f_w(x) \;\approx\; f_{w_i^\star}(x) \;+\; J_i(x)\,(w - w_i^\star). \tag{12}$$

Substituting into (2) yields

$$\widetilde{\mathcal{L}}_{\text{fn}}(w) \;=\; \sum_{i=1}^{N} \alpha_i\, (w - w_i^\star)^\top G_i\, (w - w_i^\star), \qquad G_i \;=\; \mathbb{E}_{x\sim\mathcal{D}_i}\Big[\, J_i(x)^\top J_i(x) \,\Big]. \tag{13}$$

The matrix $G_i \in \mathbb{R}^{d\times d}$ is positive semidefinite by construction and does not depend on any optimality assumption on the expert.

4.2 Closed-form solution

Expanding (13) and setting $\nabla_w \widetilde{\mathcal{L}}_{\text{fn}}(w) = 0$ gives

$$\sum_{i=1}^{N} \alpha_i G_i \left(w - w_i^\star\right) \;=\; 0 \;\iff\; \left(\sum_i \alpha_i G_i\right) w \;=\; \sum_i \alpha_i G_i w_i^\star. \tag{14}$$

When $G_{\text{agg}} = \sum_i \alpha_i G_i$ is invertible,

$$\boxed{w_{\text{fn}}^\star \;=\; \left(\sum_i \alpha_i G_i\right)^{-1} \left(\sum_i \alpha_i G_i \, w_i^\star\right).} \tag{15}$$

4.3 Specializations recover every existing scheme

Equation (15) is a template whose particular instances reproduce published methods. We enumerate the three that matter here.

1. *Linear models.* If $f_{w_i^\star}(x) = W_i^\top x$ then $J_i(x) = x^\top$ up to trivial reshaping, so $G_i = \mathbb{E}[xx^\top]$. Replacing the expectation with its empirical estimator on $\mathcal{D}_i$ gives $G_i = \frac{1}{|D_i|} X_i^\top X_i$, which up to scaling is the RegMean curvature metric.

2. *Nonlinear networks, MSE head.* Applied layer-by-layer with $X_i^{(j)}$ denoting the input activations of linear layer j when feeding $\mathcal{D}_i$ through expert i, the per-layer Jacobian has rank-one structure and $G_i^{(j)} = X_i^{(j)\top} X_i^{(j)}$ recovers RegMean in its original form.

3. *Probabilistic heads.* If the head emits a distribution $p_w(\cdot \mid x)$ and the functional metric in (2) is replaced by the symmetric KL, then G_i becomes the expected Fisher information $F_i = \mathbb{E}_{x,y}\big[\nabla_w \log p_w(y \mid x)\, \nabla_w \log p_w(y \mid x)^\top\big]$ evaluated at $w_i^\star$, and (15) recovers Fisher-weighted averaging.

Table 1 summarizes the correspondence.

Method	Curvature metric G_i	Derivation route
CRL (this paper)	$H_i = \nabla^2 L_i(w_i^\star)$	Loss Taylor, optimality assumed
HTCL realistic	$H^{(t-1)}$ with $g^{(t-1)}$ term	Loss Taylor, optimality relaxed
Fisher merging	F_i, empirical Fisher of head	Function matching under KL
RegMean	$X_i^\top X_i$ per linear layer	Function matching, linear head
Unified (this note)	$\mathbb{E}_{\mathcal{D}_i}[J_i^\top J_i]$	Function matching, Eq. (15)

Table 1: All four prior methods arise as specializations of Equation (15) under different linearization targets and distributional assumptions.

5 The Role of the Hessian

A natural reading of Sections 3 and 4 is that the function-matching formulation somehow sidesteps the Hessian in favor of the Jacobian Gram $J_i^\top J_i$. This reading is misleading. The matrix G_i that enters Equation (15) is itself a Hessian, just of a different functional. This section makes that relationship explicit, shows how the loss Hessian H_i decomposes into a Gauss-Newton piece and a residual piece, and gives the most general closed form that uses the full loss Hessian when available.

5.1 The Gauss-Newton identity

Write the per-task loss as a composition $L_i(w) = \ell_i(f_w)$ for some outer scalar loss $\ell_i : \mathbb{R}^o \to \mathbb{R}$. Standard calculus gives

$$\nabla_w L_i(w) \;=\; J_i(x)^\top \, \nabla_f \ell_i\big(f_w(x)\big), \tag{16}$$

and a second differentiation produces

$$\nabla_w^2 L_i(w) \;=\; \underbrace{J_i(x)^\top \big[\nabla_f^2 \ell_i\big(f_w(x)\big)\big] J_i(x)}_{\text{Gauss-Newton term } A_i(x)} \;+\; \underbrace{\sum_{k=1}^{o} \big[\nabla_f \ell_i\big(f_w(x)\big)\big]_k \, \nabla_w^2 f_w^{(k)}(x)}_{\text{residual term } B_i(x)}. \tag{17}$$

Taking expectation under $\mathcal{D}_i$ and evaluating at $w = w_i^\star$,

$$H_i \;=\; \mathbb{E}_{x\sim\mathcal{D}_i}\big[A_i(x) + B_i(x)\big] \;=\; A_i \;+\; B_i, \tag{18}$$

where by convention $A_i = \mathbb{E}[A_i(x)]$ and $B_i = \mathbb{E}[B_i(x)]$.

Two facts follow. First, the Gauss-Newton piece A_i is a data-dependent Jacobian Gram, reweighted by the output-space curvature $\nabla_f^2 \ell_i$. For mean-squared error $\ell_i(f) = \frac{1}{2}\|f - y\|^2$ this reweighting is the identity and $A_i = \mathbb{E}_{\mathcal{D}_i}[J_i(x)^\top J_i(x)] = G_i$. For categorical cross-entropy on a softmax head it becomes $\mathbb{E}[J_i^\top \operatorname{diag}(p_w) - p_w p_w^\top) \, J_i]$, which is the *expected Fisher* of the predictor. Second, the residual piece B_i is suppressed by the gradient-of-loss $\nabla_f \ell_i$ at the expansion point. When the expert has converged so that $\ell_i(f_{w_i^\star}) \approx \min \ell_i$, this gradient is small and B_i becomes negligible. In this regime, $H_i \approx A_i = G_i$ and the CRL formula and the function-matching formula coincide.

Proposition 1 (G_i is an exact Hessian)**.** *Define the per-expert functional objective* $Q_i(w) = \frac{1}{2}\,\mathbb{E}_{x\sim\mathcal{D}_i}\|f_w(x) - f_{w_i^\star}(x)\|^2$. *Then* $\nabla_w^2 Q_i(w)\big|_{w=w_i^\star} = G_i$ *exactly, with no approximation.*

Proof. Apply (17) with $\ell_i(f) = \frac{1}{2}\|f - f_{w_i^\star}(x)\|^2$. The output-space Hessian is $\nabla_f^2 \ell_i = I_o$, so the Gauss-Newton term is $J_i(x)^\top J_i(x)$. The residual term is weighted by $\nabla_f \ell_i = f_w - f_{w_i^\star}$, which vanishes identically at $w = w_i^\star$. The entire second derivative collapses to G_i at that point. □

Proposition 1 is the reason that the function-matching formulation does not require any residual-term approximation. It is using the true Hessian of its own objective.

5.2 Incorporating the full loss Hessian

When H_i is available in its full form (for instance computed via double backpropagation, or approximated through K-FAC or block-diagonal Fisher), we can use it directly rather than substituting G_i. The closed form mixes the loss Hessian with the functional Gram in a controllable way. Define

$$M_i(\beta) \;=\; \beta\, H_i \;+\; (1-\beta)\, G_i, \qquad \beta \in [0,1], \tag{19}$$

and the general merged weight

$$\boxed{w_\beta^\star \;=\; \left(\sum_i \alpha_i M_i(\beta) + \lambda I\right)^{-1} \left(\sum_i \alpha_i M_i(\beta)\, w_i^\star \;-\; \beta \sum_i \alpha_i g_i \;+\; \lambda\, \bar{w}\right).} \tag{20}$$

Three limits deserve attention.

- $\beta = 1$ and $g_i \neq 0$ recover the realistic loss-based formula (10) from Section 3.2. This uses the true loss Hessian and is the strongest theoretical choice when experts are close to convergence.
- $\beta = 0$ and g_i dropped recover the function-matching formula (26) (derived next in Section 6). This is the RegMean-style choice and is robust to non-convergent experts.
- Intermediate β is useful when G_i is easy to compute but we know residuals are nonnegligible. Treating β as a hyperparameter and cross-validating it is one way to capture residual curvature without paying the full cost of H_i.

5.3 Second-order expansion of the predictor

A different way to bring more curvature into the function-matching derivation is to extend the predictor Taylor expansion itself from first to second order,

$$f_w(x) \;\approx\; f_{w_i^\star}(x) \;+\; J_i(x)(w - w_i^\star) \;+\; \tfrac{1}{2}\mathcal{T}_i(x)\big[w - w_i^\star,\, w - w_i^\star\big], \tag{21}$$

where $\mathcal{T}_i(x) \in \mathbb{R}^{o\times d\times d}$ is the output-parameter Hessian tensor of f_w at $w_i^\star$. Substituting (21) into the squared-error objective (2) produces a quartic functional in $(w - w_i^\star)$ with no closed form. Stationarity gives a coupled nonlinear system whose complexity is comparable to training the merged model by gradient descent, which defeats the purpose of a dataless closed-form merge. We therefore keep the predictor expansion at first order and, if more curvature is required, switch to the Hessian-mixed formulation (20) above.

5.4 The G, J, X hierarchy

The curvature objects that appear throughout this note are organized by level of aggregation. Table 2 fixes shapes and meanings.

Symbol	Shape	Meaning
$J_i(x)$	$o \times d$	per-example predictor-to-parameter Jacobian at $w_i^\star$
$J_i(x)^\top J_i(x)$	$d \times d$	per-example Gauss-Newton kernel
G_i	$d \times d$	dataset-averaged Gauss-Newton kernel, $\mathbb{E}_{\mathcal{D}_i}[J_i^\top J_i]$
X_i	$n_i \times m$	stacked input matrix of expert i
$X_i^\top X_i$	$m \times m$	input Gram matrix
H_i	$d \times d$	loss Hessian $\nabla_w^2 L_i$ at $w_i^\star$
F_i	$d \times d$	expected Fisher information at $w_i^\star$

Table 2: Shapes and roles of the curvature objects used in this note. Here d is the parameter dimension, m is the input dimension, o is the output dimension, and n_i is the sample count for expert i.

The reason RegMean gets away with storing only the small $m \times m$ matrix $X_i^\top X_i$ rather than the large $d \times d$ matrix G_i is a Kronecker identity. For a linear layer $f_w(x) = Wx$ with $W \in \mathbb{R}^{o\times m}$ and $w = \mathrm{vec}(W) \in \mathbb{R}^{om}$,

$$J_i(x) \;=\; x^\top \otimes I_o \;\in\; \mathbb{R}^{o\times om}, \qquad J_i(x)^\top J_i(x) \;=\; (x\,x^\top) \otimes I_o \;\in\; \mathbb{R}^{om\times om}, \tag{22}$$

and summing over examples,

$$G_i \;=\; (X_i^\top X_i) \otimes I_o. \tag{23}$$

The full parameter-space Gram is therefore block-diagonal with o identical blocks, each equal to $X_i^\top X_i$. Inverting such a Kronecker structure reduces to inverting a single $m \times m$ matrix, which is the RegMean computational shortcut. Stored memory shrinks from $O(d^2) = O(o^2 m^2)$ to $O(m^2)$.

For deep networks, RegMean applies (23) layer by layer. Per-layer input Grams $X_i^{(j)\top} X_i^{(j)}$ are computed independently and substituted into (26) for each layer's weights. What this loses is the cross-layer coupling of the full G_i, which would otherwise encode how a perturbation in layer j propagates to layers $j+1, j+2, \ldots$ Dropping these off-block entries is a block-diagonal approximation of the parameter-space Gram. It is the main reason RegMean scales to transformers where full-parameter-space merging would be intractable.

Remark 2 (When to prefer G over $X^\top X$)**.** *The identity* (23) *is exact only for a single linear layer. For recurrent or attention-heavy layers, for nonlinear activations with shared parameters, or for policies with a softmax head, G_i has nontrivial structure beyond a Kronecker product. Using $X_i^\top X_i$ in these cases is a heuristic that tends to work but is not derivable from* (15)*. If the merged model is failing on such components, computing G_i directly via vector-Jacobian products on a small data sample is the principled correction.*

6 Regularized Solution for Practical Deployment

Two practical concerns complicate the direct use of (15) in deep networks. First, G_{agg} is typically rank-deficient in directions unobserved by any dataset, so direct inversion is unstable. Second, when experts are far apart in parameter space, the linearization $f_w \approx f_{w_i^\star} + J_i(w - w_i^\star)$ is only locally accurate and the argmin of (13) can land in a region where no J_i is trustworthy.

We address both concerns with a Tikhonov-regularized surrogate that anchors the solution to the α-weighted ensemble mean $\bar{w} = \sum_i \alpha_i w_i^\star$,

$$\widetilde{\mathcal{L}}_{\text{fn}}^{\lambda}(w) \;=\; \widetilde{\mathcal{L}}_{\text{fn}}(w) \;+\; \tfrac{\lambda}{2}\,\|w - \bar{w}\|^2, \qquad \lambda > 0. \tag{24}$$

Stationarity gives

$$\left(\sum_i \alpha_i G_i + \lambda I\right) w \;=\; \sum_i \alpha_i G_i w_i^\star + \lambda \bar{w}, \tag{25}$$

and therefore the regularized merged weight is

$$\boxed{w_{\text{fn},\lambda}^\star \;=\; \left(\sum_i \alpha_i G_i + \lambda I\right)^{-1} \left(\sum_i \alpha_i G_i\, w_i^\star + \lambda\, \bar{w}\right).} \tag{26}$$

If the experts are not stationary and their gradients $g_i = \nabla L_i(w_i^\star)$ are available, the loss-based correction from (10) combines with (26) to give the most general form we consider,

$$\boxed{w_{\text{gen}}^\star \;=\; \left(\sum_i \alpha_i G_i + \lambda I\right)^{-1} \left(\sum_i \alpha_i G_i\, w_i^\star - \sum_i \alpha_i g_i + \lambda\, \bar{w}\right).} \tag{27}$$

Two limiting regimes are worth noting.

- *Small λ.* Equation (27) reduces to (15) modified by the gradient correction. The curvature metric dictates how expert optima are blended.

- *Large* λ. Equation (27) tends to $\bar{w}$. This is the safe fallback when all curvature estimates are unreliable or when experts are so far apart that no local linearization applies.

7 Application to Reinforcement Learning Model Merging

We return to the question that motivated this note: can the HTCL methodology be used to merge expert policies trained on different tasks in a reinforcement learning setting?

7.1 Why the original CRL formula fails in RL

The two failure modes the user observed empirically correspond to two identifiable theoretical violations.

1. *Non-stationary experts.* In RL, actor and critic losses are computed against non-stationary targets (bootstrap returns, target networks, changing exploration distributions). Even after many updates, $\nabla L_i(w_i^\star)$ is a small but nonzero random variable, not identically zero. Assumption 1 therefore fails and the dropped gradient term in (7) becomes a first-order error.

2. *Distant expert optima.* Policies trained on different RL tasks typically do not share loss-landscape basins. The Taylor expansion around each $w_i^\star$ has a validity radius that is smaller than the inter-expert distance, so the surrogate (4) is poorly calibrated at the argmin of $\widetilde{\mathcal{L}}_{\text{loss}}$.

7.2 Proposed fixes

Fix 1: switch to the function-matching objective. Equation (26) requires neither $\nabla L_i(w_i^\star) = 0$ nor access to H_i. For actor networks producing distributions $\pi_{w_i^\star}(\cdot \mid s)$, replacing MSE with KL in (2) and linearizing the log-density yields

$$G_i \;=\; \mathbb{E}_{s\sim\rho_i}[\, F_i(s)\,]\,, \qquad F_i(s) = \mathbb{E}_{a\sim\pi_{w_i^\star}(\cdot|s)}\Big[\nabla_w \log \pi_{w_i^\star}(a|s)\, \nabla_w \log \pi_{w_i^\star}(a|s)^\top\Big]\,, \tag{28}$$

where ρ_i is the state-visitation distribution of expert i. This is the *policy Fisher information*, which is the correct curvature for KL-merging of stochastic policies.

Fix 2: keep the gradient term. When computing G_i via back-propagation on $\mathcal{D}_i$, the gradient $g_i = \nabla L_i(w_i^\star)$ can be obtained at essentially no additional cost. Use Equation (27) rather than (26).

Fix 3: mitigate the distant-optima problem. Equation (27) handles this only through the λ term. Three algorithmic remedies are complementary.

1. *Layer-wise merging.* Apply (27) per linear layer using the layer-input Gram matrix as $G_i^{(j)}$. The linearization radius per layer is substantially larger than in the full-parameter formulation. This is the trick RegMean uses to scale from linear regression to transformers, and it transfers to RL actor and critic heads.

2. *Iterative re-linearization.* Perform K rounds of (27) where each round recomputes J_i (and optionally g_i) at the *current* merged weight rather than at $w_i^\star$. This is the proximal-point or sequential-quadratic-programming analogue of HTCL's hierarchy, applied in parallel rather than sequentially.

3. *Rebasin before merging.* When the experts have the same architecture, permutation-alignment (weight matching, activation matching) can be used to bring $w_i^\star$ into a common basin before applying (27).

7.3 Will it work?

The honest answer is that (27) is strictly better founded than the CRL formula you attempted previously, because it does not require Assumption 1 and it uses a curvature metric that is cheap to compute for deep networks. It will outperform naive averaging and vanilla Fisher-weighted averaging whenever the expert predictors are approximately linear on a neighborhood containing the merged weight. For policies trained on tasks whose optima lie in different loss basins, closed-form merging alone is insufficient, and the three algorithmic remedies above should be combined with it. The iterative re-linearization path is the closest analogue to HTCL's hierarchy and is likely the most effective in practice.

8 Summary

We derived a single framework that reproduces CRL, HTCL, Fisher-weighted averaging, and RegMean as specializations of the same quadratic surrogate. The crucial object is a per-expert curvature metric. The general closed-form merged weight is

$$w^\star \;=\; \left(\sum_i \alpha_i M_i + \lambda I\right)^{-1} \left(\sum_i \alpha_i M_i\, w_i^\star \;-\; \beta \sum_i \alpha_i g_i \;+\; \lambda\, \bar{w}\right), \tag{29}$$

with $M_i = \beta H_i + (1-\beta)\, G_i$ interpolating between the loss Hessian and the Gauss-Newton kernel. Setting $\beta = 0$ recovers the RegMean-style functional merge, $\beta = 1$ with g_i retained recovers the realistic CRL formula and the HTCL update, and intermediate β trades off curvature fidelity against computational cost. With the Gauss-Newton choice $G_i = \mathbb{E}_{\mathcal{D}_i}[J_i^\top J_i]$, no assumption on expert optimality is required, the formula degenerates gracefully to ensemble averaging as $\lambda \to \infty$, and it recovers RegMean, Fisher merging, and CRL as special cases. The key conceptual point is that the Hessian is never discarded: either it appears as the true loss Hessian H_i through the decomposition $H_i = G_i + B_i$, or it appears as the exact Hessian of the functional surrogate via Proposition 1.